



SUPPLY CHAIN & SALES ANALYTICS · B.A.D FRAMEWORK DIAGNOSTIC

Late Delivery at 45.8% & Returns at 5.9%

Where's the Root Cause?

A diagnostic of logistics operations, quantifying the profit impact across the full portfolio.

5,009

orders analyzed

45.8%

late delivery rate

2

independent root causes

Ship Mode and Seasonality are two independent operational drivers behind the gap to target, while the Return Rate is eroding \$20,435 in profit

58.1%

Late rate, First Class

vs 2.9% Same Day · Spread 55.2 pp

76.6%

Late rate, February (peak)

vs 23.7% November · Spread 52.9 pp

\$20,435

Lost Profit from Returns

Technology leads at \$11,379 (55.7%)



THE TAKEAWAY

Late Delivery Rate at 45.8% exceeds the 30% target (+15.8 pp), driven mainly by two independent operational root causes — Ship Mode routing and Q1 seasonal capacity — while the Return Rate at 5.9% (+0.9 pp over target) is a separate product-quality issue, concentrated in Machines & Technology.

We break the problem into testable branches, then validate each one with DAX across the full model

Objective: (1) identify the root cause, (2) propose a recovery path — via a MECE issue tree (B.A.D)



Each branch is a hypothesis: ✓ confirmed / ✗ rejected. Region and Segment were ruled out because their spread is too small relative to Ship Mode and Seasonality.

Across all 5,009 orders, revenue is stable but operations are far off target

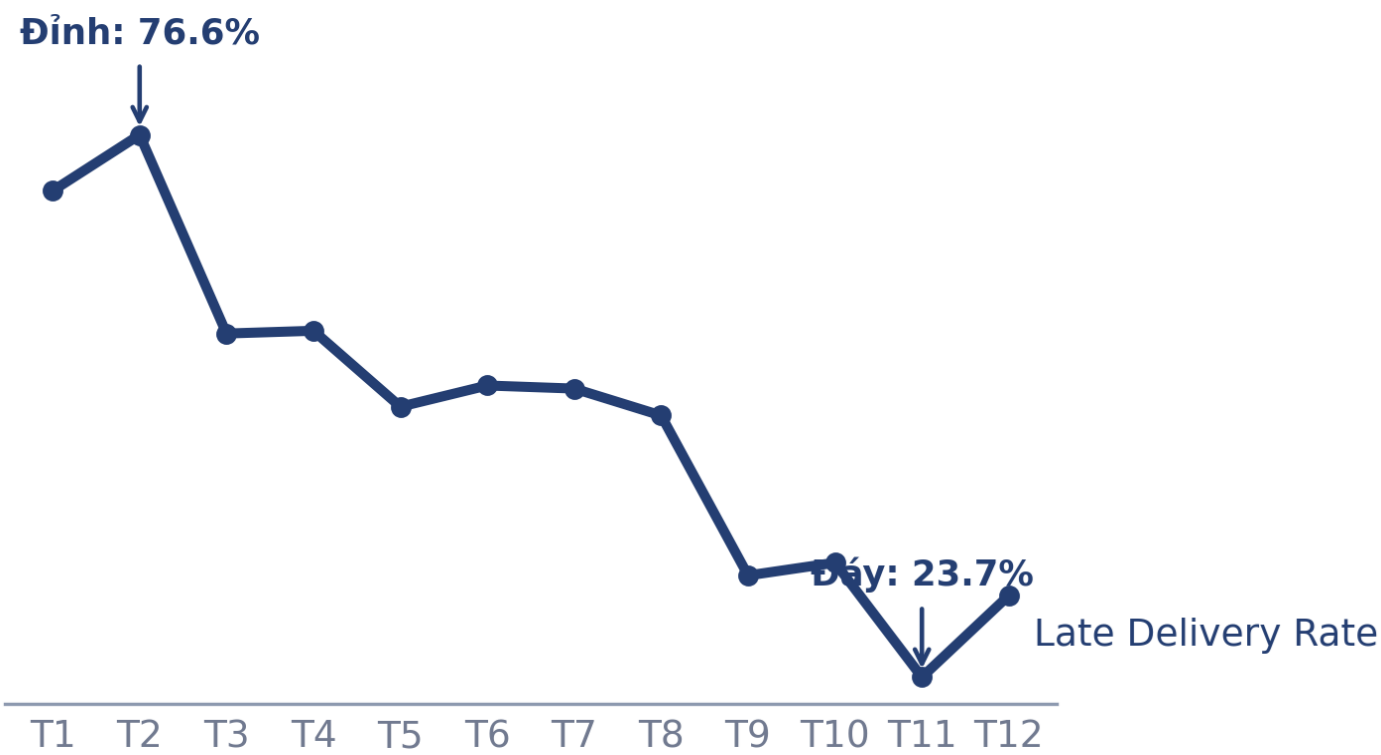
\$2,297,201 Total Revenue

12.5% Profit Margin (Target 15%)

54.2% On-Time Delivery Rate

34.6 Avg Delivery Days

Late Delivery Rate by month — ranging from 23.7% to 76.6% over the year

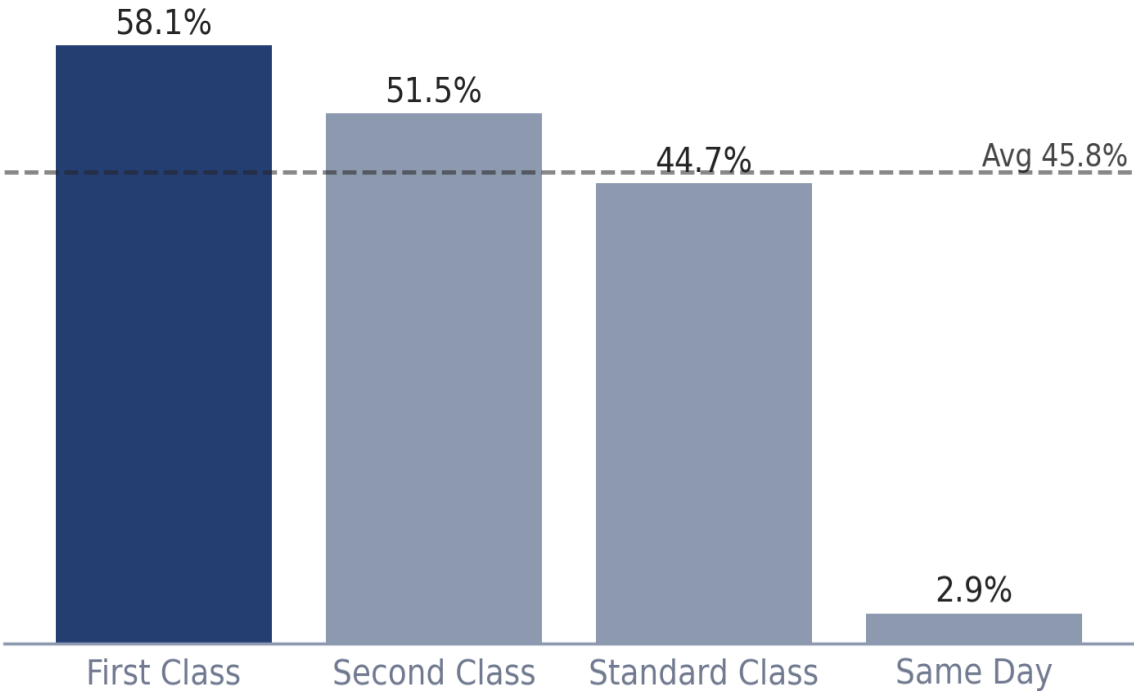


COMPLICATION

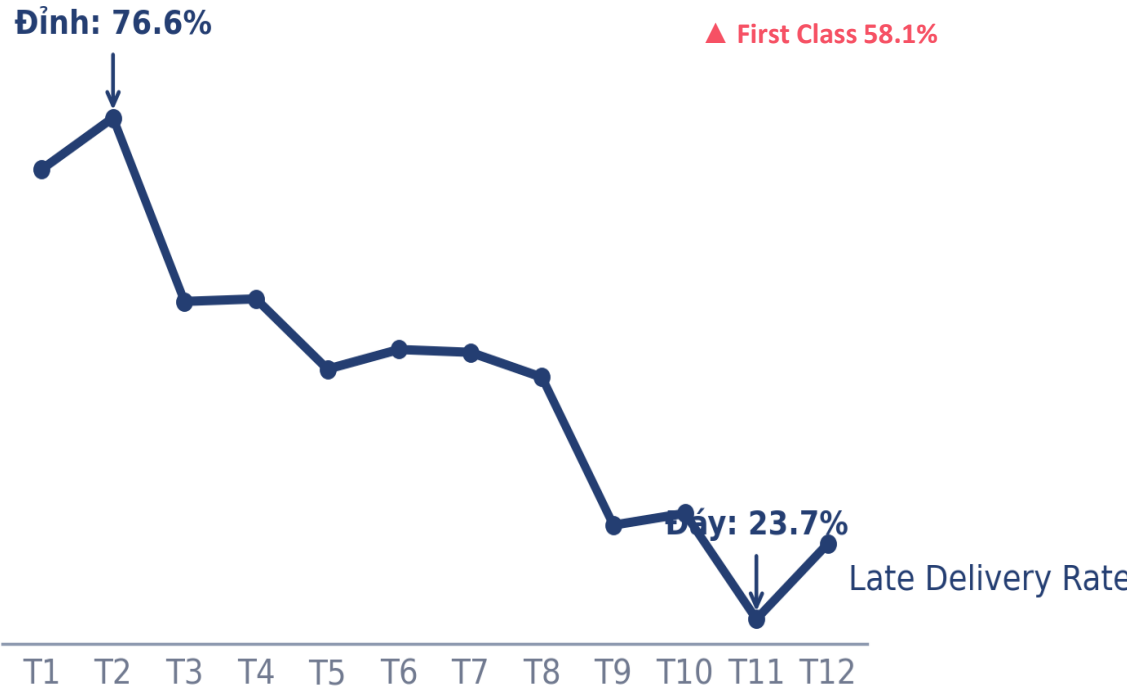
The fastest shipping service is late most often, and February runs 3.2x later than November

Question: is this one problem or two independent causes, and where should we prioritize first?

Late Delivery Rate by Ship Mode



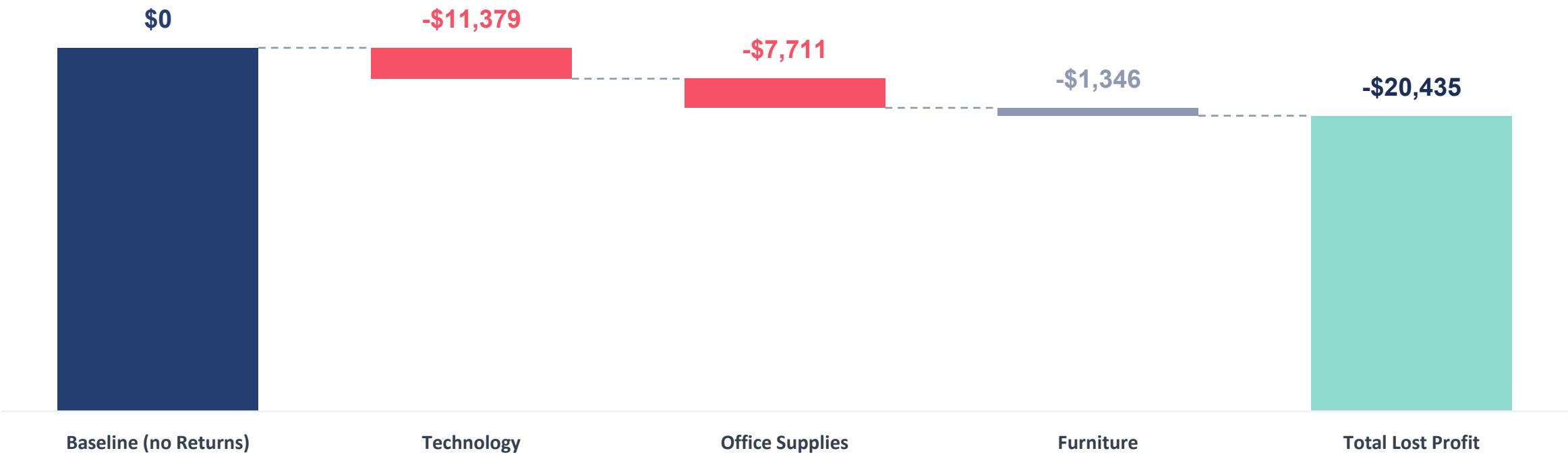
Late Delivery Rate by Month



The average Late Delivery Rate of 45.8% hides two distinct patterns: First/Second Class run ~18–20x later than Same Day, and January–February are more than 3x later than November — while Region and Segment are nearly uniform.

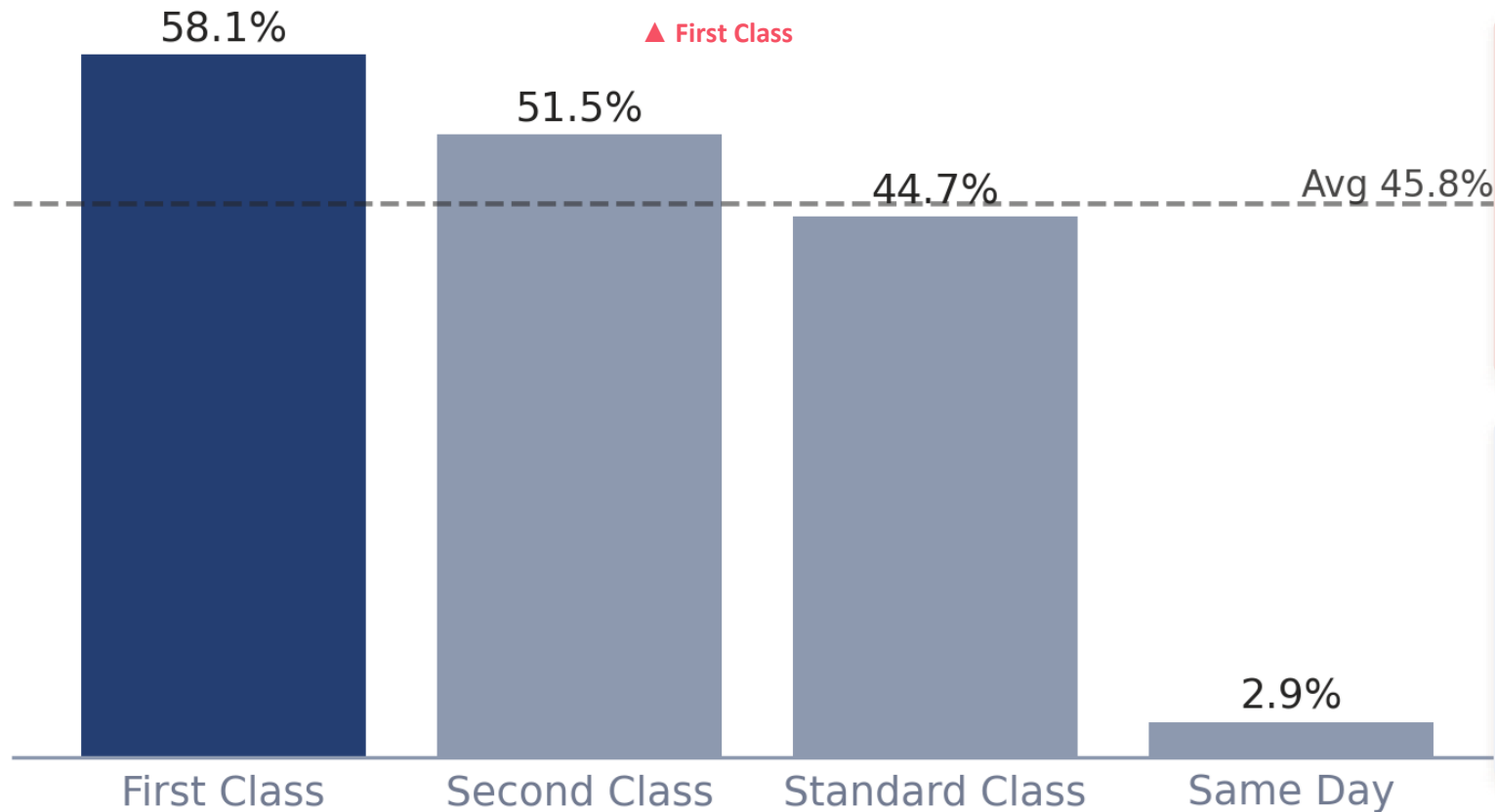
DECOMPOSITION

\$20,435 in Lost Profit from Returns, broken into 3 non-overlapping categories



Formula: Returned orders × Avg profit per successful order, by category. Technology accounts for 55.7% of total losses despite only 21.9% of total orders — the highest-value place to intervene on quality.

First Class & Second Class run 18–20x later than Same Day — the paradox of the fastest service being the slowest



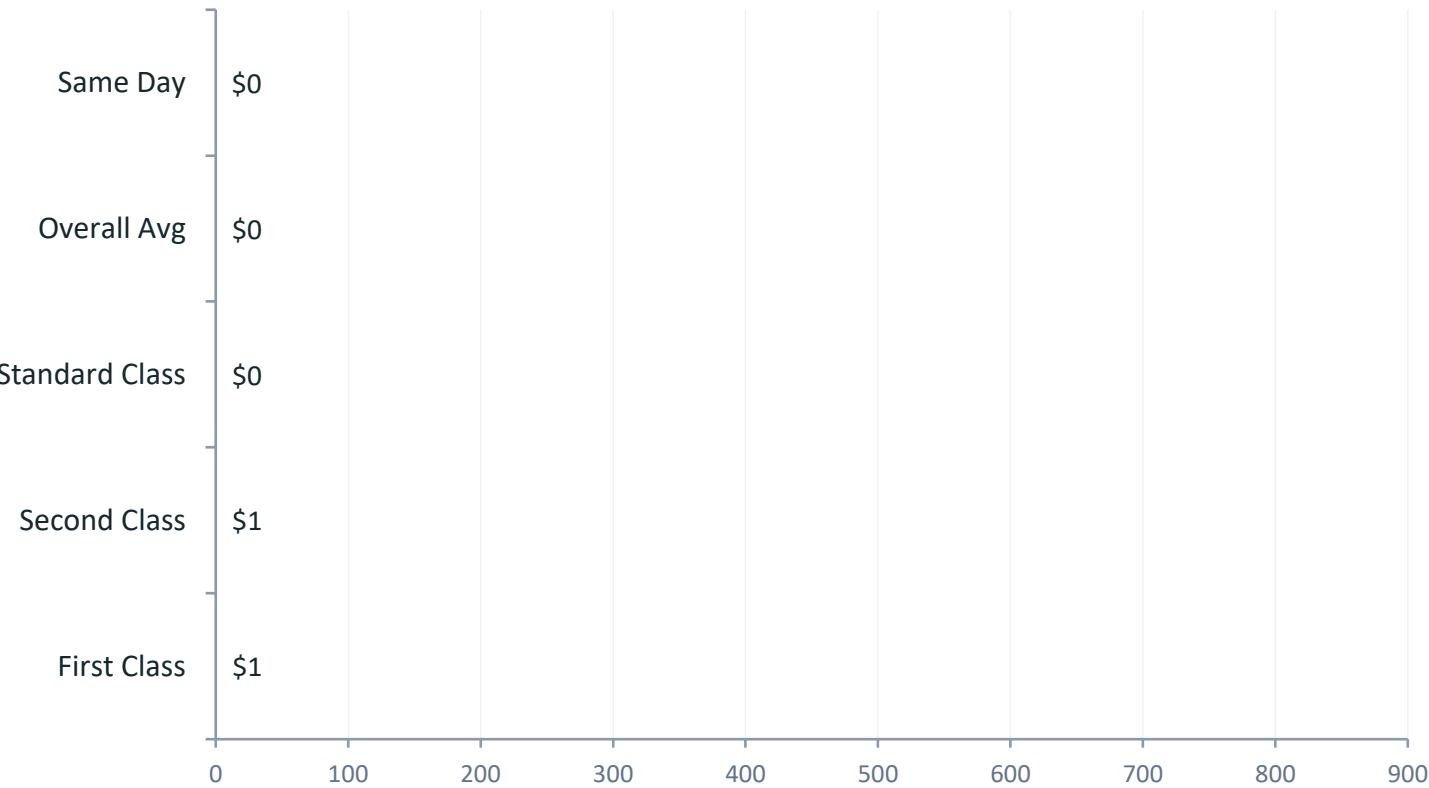
NOT EXPLAINED BY DISTANCE OR SMALL SAMPLE SIZE

Same Day is late only 2.9% of the time (7/264 orders), while First Class is late 58.1% (462/787 orders) and Second Class 51.5% (480/964 orders) — a gap too large to explain by random variation.

LARGE ENOUGH IN SCALE TO BE A SYSTEMIC ISSUE

First + Second Class account for 1,751/5,009 orders (35%) but contribute 942/2,266 late orders (41.6% of all late orders) — not localized risk, but a routing/carrier process failure.

A 55.2 pp spread across Ship Mode is the strongest driver among the 5 MECE hypotheses tested



55.2 pp

Late Rate spread: First Class (58.1%) – Same Day (2.9%)

LATE RATE BY SHIP MODE (VS. SAME DAY)

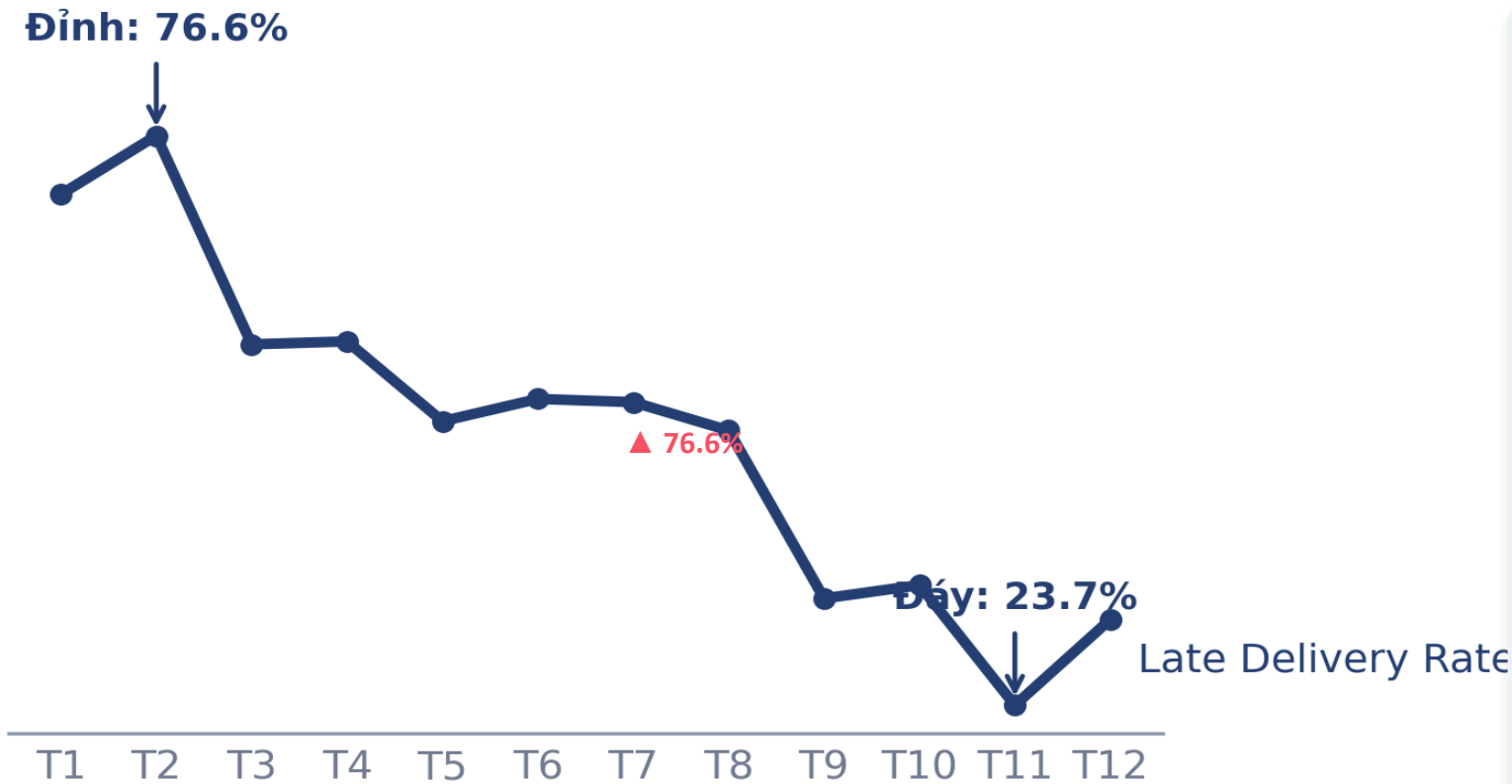
First Class	×19.9
Second Class	×17.6
Standard Class	×15.3
Overall Avg	×15.8
Same Day	×1.0

Late Delivery and Returns are both evenly distributed across Region & Segment — not a localized issue



So what: Late Delivery Rate ranges narrowly from 44.5%–48.2% across 4 regions (spread 3.7 pp), and Return Rate ranges from 5.4%–6.1% across 3 segments (spread 0.8 pp) — confirming the root cause is not geography or customer behavior, but operations (Ship Mode) and timing (seasonality).

February is the peak of the logistics crunch, with a Late Rate 3.2x that of November



A STEADY, DECLINING TREND — NOT RANDOM NOISE

Late Delivery Rate peaks in February (76.6%, 197/268 orders), stays elevated through Q1–Q2 (>50%), then declines steadily through Q3–Q4 to its low in November (23.7%). This recurring, monotonic monthly pattern points to a seasonal operating-capacity issue (early-year peak), not random variation.

→ Late Delivery Rate should be tracked monthly, not just as an annual average.

The evidence points to two independent, compounding operational drivers — separate from the product-quality issue



THE TRIGGER

First/Second Class routing process is out of standard

Explains 41.6% of all late orders despite only 35% of volume — not due to distance or carrier speed, but how these two Ship Modes are operated internally.



THE AMPLIFIER

Q1 seasonal pressure amplifies the Ship Mode issue

January–February Late Delivery Rate is double the annual average — First/Second Class orders placed in this window face a compounded risk of delay (routing failure + seasonal overload).

What the data cannot yet confirm: whether the root cause of Ship Mode routing sits with a third-party carrier or MAD's internal process — warehouse/carrier operating logs are needed to confirm, so we flag this rather than overstate it.

RECOMMENDATIONS

3 actions, 2 operational levers: recover ~942 Ship Mode late orders and narrow the 52.9 pp seasonal spread



01

Audit and restructure the First & Second Class shipping process

Review the routing, carrier, and intake process for First/Second Class; benchmark against Same Day/Standard to find bottlenecks.

Potential to reduce ~942 late orders (41.6% of 2,266 total late orders)

↳ Addresses RC1 · Ship Mode Paradox



02

Deploy an operational capacity buffer for the January–February peak season

Scale up staffing/warehouse capacity ahead of Q1 based on the recurring annual Late Rate pattern; establish a dedicated SLA for the peak period.

Narrow the seasonal spread from 52.9 pp to near the 30% target

↳ Addresses RC2 · Seasonality Shock



03

Run quality control (QC) and a supplier review for the Machines & Technology lines

Sample-test Return reasons for Machines (11.6%) and Technology (8.0%); distinguish genuine product defects from mis-categorized returns.

Recover up to \$11,379 in lost profit from Technology (55.7% of total losses)

↳ Addresses Return Rate · Product Quality

Tackle the high-impact quick win first, then sequence the two larger initiatives



1

Now — Quick win

Audit the First/Second Class process. High impact, low effort.

3

45–60 days

QC the Machines/Technology lines. High impact, medium effort.

2

60 days (ahead of next Q1)

Seasonal capacity buffer. High impact, heaviest effort.

Lock in alert thresholds so operational issues don't quietly recur



Alert at 35%

Target + 5 pp, proactively alert the owner, root-cause within 48h

Escalate at 40%

Cross-functional review; prioritize fixing Ship Mode/Seasonality immediately

Watchlist KPIs

Late Delivery Rate % (by Ship Mode & Month), Return Rate % (by Sub-Category), Profit Margin vs Target



MAD SUPPLY CHAIN

Thank you

Two independent operational root causes, diagnosed and both fully addressable.

Questions welcome — methodology & DAX query working papers are in the appendix.